Design for Testing

Handbook

SIERRA CIRCUITS

Scope: DFT checks all the aspects of a PCB design and pinpoints hidden defects, incorrect arrangements of components, and other faults, if present. This Design for Testing Handbook conveys the fundamental idea of various testing techniques, and their benefits and drawbacks.

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FLYING PROBE TESTING

SEICA TESTING CAPABILITES

- Performs Open and Short Circuit testing
- Measures component values. Resistor values of .1Ω – 100MΩ, Caps value from 1pF -100mF and Inductance values from 1uH -10H can be checked
- Tests component placement and polarity
- Identifies missing components

BETTER TESTING

BETTER BOARDS

Testing a PCB after manufacturing plays a pivotal role in procuring a flawless design. Design for testing (DFT) evaluates the board's accuracy based on functionality and manufacturability. It not only ensures a perfect design but also cuts down the manufacturing cost, substantially.

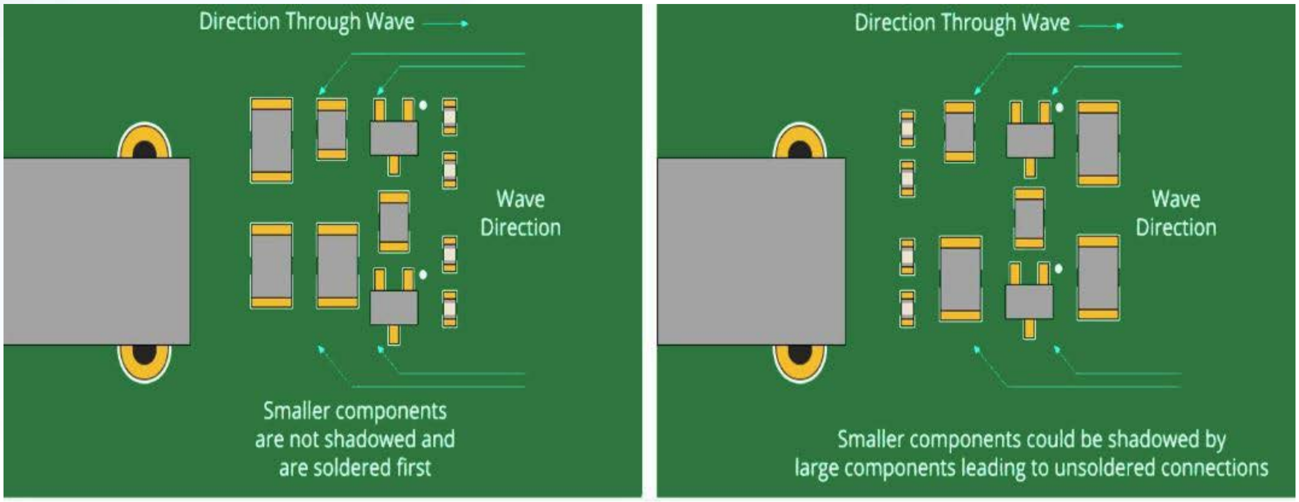
1. Design for testing fundamentals

DFT is a method of operational and functional testing of a circuit board. This methodology identifies any short, open circuit, wrong placement of the components, or faulty components. Design for testing incorporates the addition of some test points on the PCB depending on the design requirements and the parameters of interest. This testing method is performed to validate three major questions:

- **Is the board designed precisely?**
- **Is the board fabricated flawlessly?**
- **Do all the components, ICs, and connections operate perfectly?**

Apart from these, the other major issues considered are:

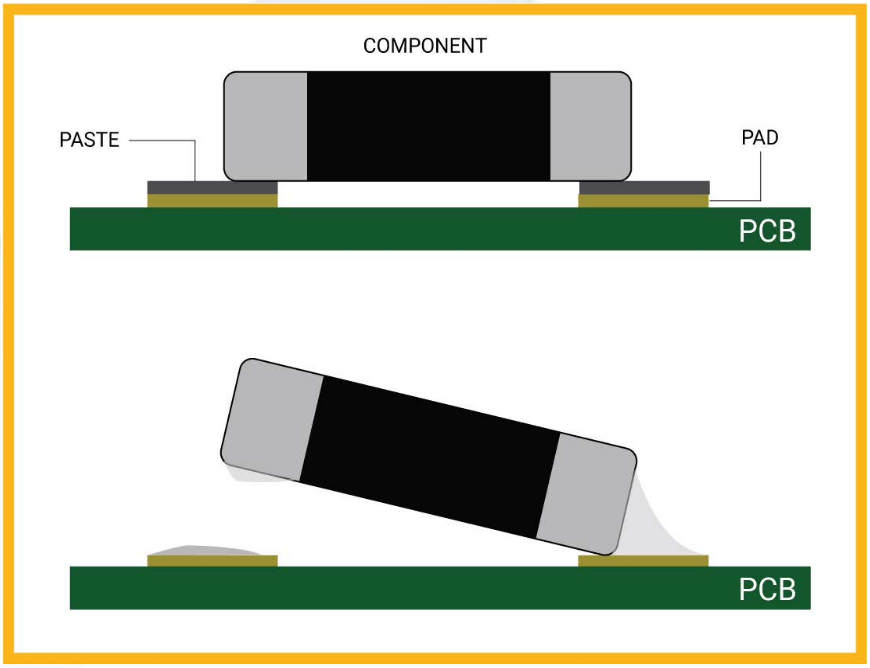
- **Component misplacement:** One of the key aspects of board design is where and how a component is to be mounted on the board. For example, placing taller components in front of shorter ones causes serious solder joint issues.



Left: Correct positioning of components.
Right: Incorrect positioning of components

Inaccurate component clearance: Appropriate gap between a component to another and a component to the board edges is crucial. Close spacing will increase the risk of a defective design.

Improper surface-mount pad size: It is a typical design problem. Improper pad sizes result in tombstoning. Here, passive SMT components such as capacitors, and resistors get lifted off from their pad. This is commonly known as the tombstone effect.



Top: Component strongly soldered on a PCB.
Bottom: Tombstone defect

- **Excess or less solder mask:** If the solder mask between pads is inadequate, the electrical connections can degrade.
- **Wrong or faulty electrical components:** Determination and rectification of wrong and defective components are critical before the designs are ready for production. If it remains unnoticed, it can turn out to be an expensive mistake.

To check the manufacturability of your circuit design, try our Better DFM tool. This will help you in simplifying the product design and enhances the functional performance of your board.

2. Importance of DFT

In the earlier days, the number of components was hardly around 100 to 200. Hence, the board had enough room to place test points. Now, the entire fabrication technology has gone through a revolutionary change. HDI PCB designs include thousands of components and solder connections. The space constraint becomes a concerning issue in deciding proper test positions on a dense board. If any component or connector hampers the design, it would be a nightmare for fabricators and designers.

DFT engineers and product developers establish a set of testing methods to find out any inaccuracies and produce a high-quality circuit board.

3. Parameters to include in DFT

Here are some of the parameters that you can include in board testing to ensure efficiency and precision.

- **Test points:** Test point or fixture insertion is a requisite technique in DFT to increase test efficiency. The position of a test point is decided based on how many components it can cover. The fixtures are distributed strategically in such a way that they can easily access different parts on the board. Signal integrity issues can be mitigated by arranging accurate power and ground test points. **Typically, the dimensions for the bottom-access pad used for flying probe and bed-of-nails tests are 40 mils and above.**

- **Test traces:** You can position test points on the traces that imitate the sensitive traces. These test points can be connected to oscilloscopes, TDRs, or signal generators to know the behavior of the signals. A test point can be placed in the auxiliary clock output for trigger or synchronization during testing.

- **Inclusion of LEDs:** Incorporate LEDs in testing methods to determine whether the power is switched on or off. Debug LEDs are a suitable choice for FPGA or microcontrollers since they require debugging errors in the code.

- **Selection of the test method:** Choose the method that best suits your design. Flying probe test is preferable for small production because of its simple setup and slow testing speed. In-circuit test (ICT) involves fixed programming and is suitable for large production.

- **Headers:** These are a kind of test point connected to vias to measure the voltages across it.

- **Additional circuit features:** If the board has enough room, some circuitries are introduced to check the voltage and current of the components. These are non-essential yet helpful in validating the components' rating.

4. PCB testing strategies

4.1. Border edge clearance

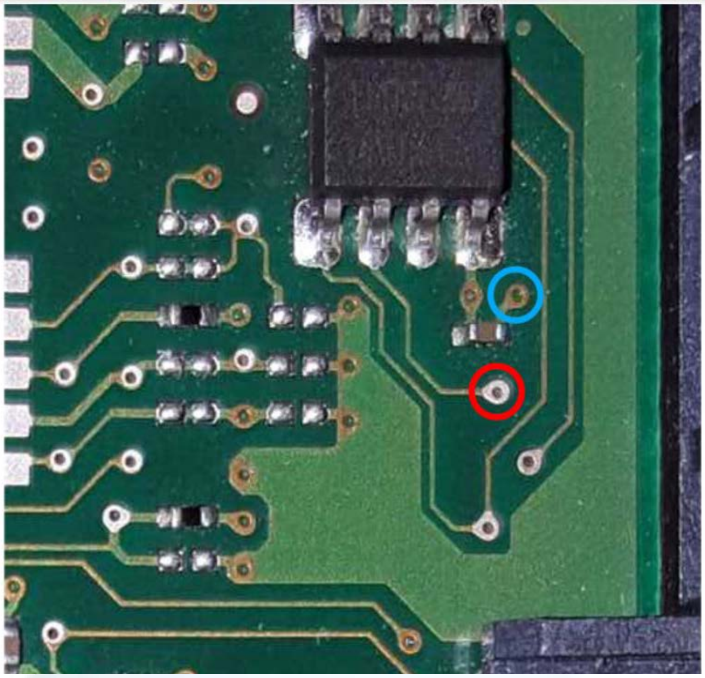
This is essential to keep sufficient, unoccupied space along the opposite edges of the board. Border edge clearances are helpful in holding the testing machine, perfectly. **Generally, the standard width of the unavailable edges is maintained at 3 mm**. Often, panel wastes are also implemented for the same purpose.

4.2. Fiducials

The testing machines need some reference points to know the exact position of the probes. The reference points, known as fiducials, are located on the panel waste, and on the PCB itself, if the waste has been removed. The most suitable and recommended fiducials are top-left and bottom-right of the board.

4.3. Vias

The vias incorporated in the board design need to be non-masked to position a probe on the edge of them.



Blue: Masked via Red: Non-masked via.
Image credit:electronics.stackexchange.

4.4 Probes near component legs

Probing near the component legs call for sufficient area on the toe. This helps in establishing a good solder joint. Having said that, probing right on top of the component legs can lead to errors since any potential open circuit could be temporarily corrected at the test point due to the pressure exerted by the probe, causing the leg to stick to the pad.

4.5. Size of the board

Place the test access points as close together as possible. It is particularly applicable for designing a large board.

4.6. Cleaning the assembly

Cleaning the assembly is essential to perform probing. This ensures the removal of unwanted flux. If the probe position is repeatedly moved to establish a strong contact it will result in increased test time and false fails.

4.7. Probe points on the ground and power planes

Keep probe points on the ground and power planes on the bottom side of a board. This allows fixed probes to act as temporary fixtures that expedite the detection of shorts. Further, the overall test time can be reduced.

4.8. Maximizing test accessibility

Increase the test accessibility on one side of the assembly. Insert one probable point for each network. Testing both sides of the board can increase the cost since the PCB needs to be turned over to test the other side. This can be neglected if a double-sided testing machine is used.

4.9. Checking the component height

There is a maximum component height allowed for both the probed and the unprobed region of a PCB, typically around **40 mm** and **90 mm**, respectively. The tall components are placed on the bottom side if it is unprobed or fitted after the test. They can create a no-fly zone on the assembly and also hinder test access.

5. Types of board testing

Here are the two important testing techniques:

5.1. Bare board testing

Bare board testing is performed to check the PCB's connectivity before assembling the components. Following are the two ways to perform this kind of testing:

- The isolation test verifies the resistance between two electrical connections.
- The continuity test checks if there is any presence of an open circuit within the board.

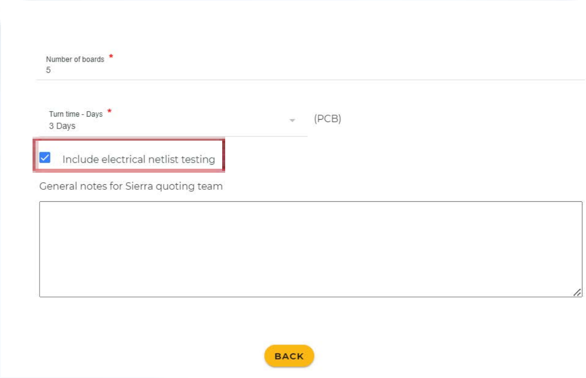
5.2. Assembled board testing

Assembled board testing is executed after assembling the components. This process ensures the circuit board's integrity and correct functionality of the components.

6. PCB E-test

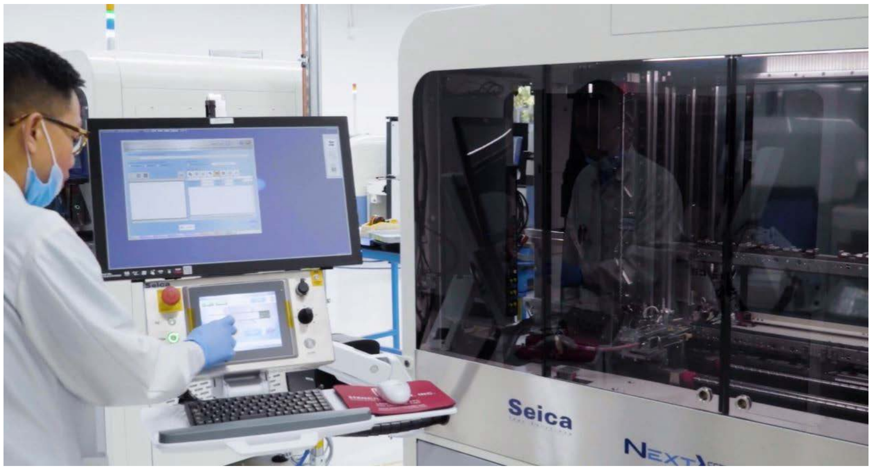
E-test examines the electrical conductivity of the circuit board with respect to the netlist file obtained from the schematic. E-tests can be carried out using two methods, flying probe and fixture method. The criteria for choosing these methods vary as per requirements.

In the Sierra Circuits' custom quote tool, you can include electrical testing for your board while requesting an estimation. Click on the check box shown below. If you have any specific testing requirements, you can mention the same in the general notes section. Time taken to perform the test can vary depending on the number of nets to be tested.



6.1. Flying probe testing

The bare board and assembled board both can incorporate the flying probe test (FPT) in passive and active mode, respectively. The probes comprise needles for checking. The test points can include passive components like resistors, capacitors, inductors, untented vias, or terminating ends of the components. It can detect the value of non-powered elements, and open or short circuits, measure voltages, and check the placement of diodes and transistors. The time taken to finish FPT depends on the number of nets to be tested. Usually, for a normal prototype board, it can take 30 mins to 3 hours.



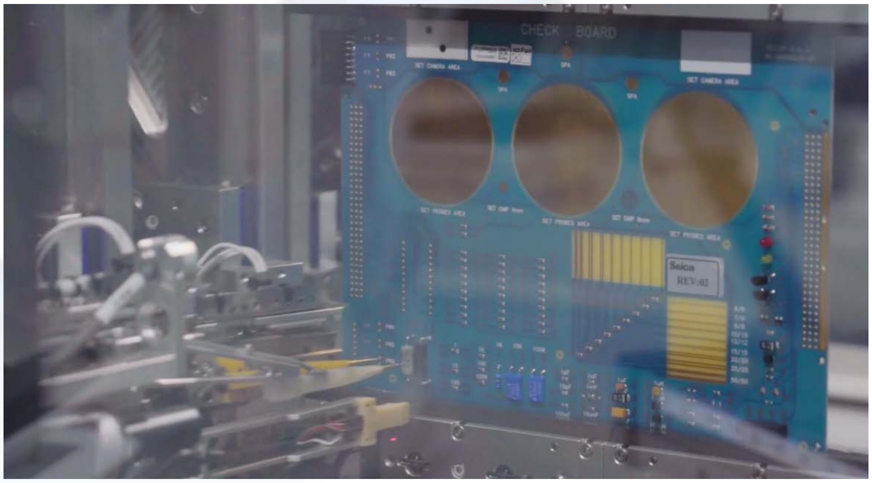
Expert working in FPT machine

6.1.1. FPT procedure

1. At first, the test program is generated offline by the test program-generating application that runs on a PC. FPT requires the BOM, ECAD, and ODB++ files. The generated FPT test program is now loaded into the tester.

2. The assembled or bare board is placed on a conveyer belt so that it can travel inside the tester area where the probes are.

3. During the test program, the probes will apply electrical test signals and power to the test points and detect shorts, opens, and wrong component placement.



Vertical testing in FPT

4. Equipped camera in FPT inspects component polarity and performs diode impedance test on inputs of integrated devices. Though, the integrated device's functional behavior is outside the testing scope.

At Sierra Circuits, we use Seica Spa Pilot V8 for flying probe testing. This latest machine provides maximum performance, increased test speed, and low to medium volume run. Simultaneous vertical probing on both sides of the circuit board quickens the debugging process and enhances the flexibility for prototyping.

Seica Spa Pilot V8 FPT machine. Image credit: Seica

6.1.2. Guidelines for test point design and placement

6.1.2.1. Accessibility to nets

- Nets linked to the active components must be accessible on the same side of the test probes. This leads to enabling the “openfix” option. Apply the same rule for SMD components as well.
- For better efficacy, maintain a distance of 2.8 mm between the probe and the probe sensor.
- This condition does not apply to the vertical testing procedure and to the board that is to be tested on both sides.

(6.1.2.1. to 6.1.2.9, source: DFT (Design for Testability) for SEICA systems)

6.1.2.2. Features of test points

- **Preferrable contact points of the FPT process are:**
  - Test pads without solder resist
  - Vias
  - Through-holes
  - PTH and SMD component pads

- **Size of a test point:**
  - Dimensions depend on the probe type
  - The minimum size is 6 mils (20 mils recommended)
  - The spacing between one test point to another should be minimum of 10 mils and 20 mils

- **Vias:**
  - The diameter should be between 8 to 20 mils (10 mils recommended)
  - Vias must be occluded and free from the solder resist

- For SMD components, particularly with steps slower than 25 mils, apply a solder pad, that is longer than the component pin. This enables the testing probe to contact the test point not on the pin but on the soldering pad itself

- For horizontal architecture (PILOT L4/H4), components should be 4 to 5 cm higher than the opposite of the test side

6.1.2.3. Board clamping

- The board is secured with a clamp/conveyed via, either horizontally (Pilot L4) or vertically (Pilot V8)
- For testers with horizontal architecture, provide support after every 150 mm or the distance based on your design requirement on both axes (except the Aerial and V8 test systems) to keep the board flat

6.1.2.4. Board dimensions

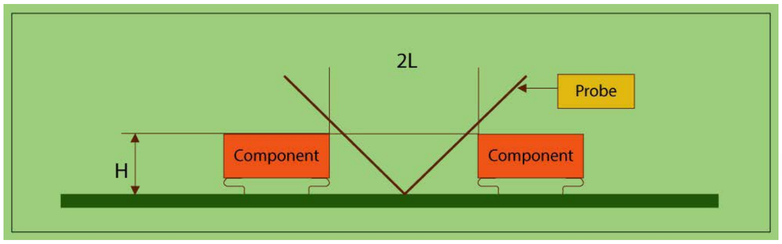
- Maximum board dimensions of 600 x 1010 mm for Pilot L4 testers
- Maximum board dimensions of 600 x 540 mm for Pilot V8 testers

6.1.2.5. Maximum component height

- CAD data must include the component height
- **Component height calculations for horizontal Pilot L4 testers:**
  Maximum board height + components on the head side: 40 mm including the board width
  Maximum board height + components on the bottom side: 100 mm
- **Component height calculations for vertical Pilot V8 testers:**
  Maximum board height + components on either side: 40 mm including the board width
- Design team provides a list of components whose height is over 17 mm

6.1.2.6. Test point distances

- The clearance is the distance between the center of the test point and the side of the components
- The minimum distance between a test point and the component can be estimated by the formula: L = (0.29 x Height) + 0.7 (in mm)
  E.g. For height H= 4 mm, L= (0,29 x 4)+ 0,7 = 1,86 mm
- GND access is provided preferably at the four angles of the board



Distance between test point to component

6.1.2.7 Fiducial recognition

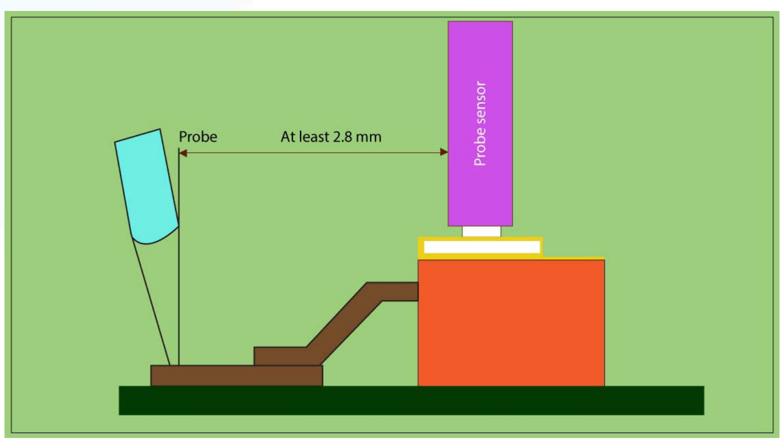
- Include 3 fiducials for each board
- Position the fiducials in a triangle shape on either side of the board. It is essential to achieve different arrangements of the fiducials to eliminate side errors
- For the larger boards with dimensions greater than 400 mm, place local fiducials concerning the fine-pitch components which must be visible after the component has been mounted

6.1.2.8 Wire connection

Wiring should be refrained from colliding with the test points

6.1.2.9 Option for openfix

- To avail openfix, nets to be tested and connected to the active components, must be accessible on the same side of the testing probe. This is applicable to SMD connectors also
- For better results, maintain a distance of 2.8 mm between the probe and the probe sensor



2.8 mm distance is maintained between the probe and the probe sensor

6.1.3. FPT benefits

- Cost-effective
- Greater test coverage
- Do not require a fixture
- Quick implementation

6.1.4. FPT drawbacks

- Time-consuming since probes move between measurement points
- Tough to set up if the board does not include any test point, test via, or masked via
- The lump-sum capacitance can only be tested for the capacitors in parallel

6.1.5. What FPT checks and verifies

- Performance of the components
- Flaws in assembly
- Open traces
- Short connections
- Misplacement of components
- Value of capacitance, resistance, and inductance

6.2 In-circuit testing

In-circuit test is also known as the bed of nails testing method. This process incorporates some pre-mounted, electrical probes aligned under the board through the preset access points. The accurate, and stable electrical connection between the probes and the PCB can be established in this way. The testing probes cause the current to flow through the predetermined test points.

ICT can check for shorts or open circuits, solder mask shortcomings, misplacement or absence of components, etc. This method comprises testing fixtures to hold the board with the probes correctly, and test jigs to check multiple components on a board, simultaneously. This test method saves time.

6.2.1. Directives to make your board ICT compatible

6.2.1.1 Accessibility to nets

As a general rule, all the nets must be accessible on the same side to the measuring heads, preferably the flip side, comprising the maximum number of components. A limited number of access points is permissible on the opposite side to eliminate the hindrance of accessibility to test pads and to increase the test coverage. This principle is not applicable to Aerial or V8 test systems.

(6.2.1.1 to 6.2.1.18, source: DFT (Design for Testability) for SEICA systems)

6.2.1.2 Accessibility to nets

- The NC nets, which are fill-in pins, need not be accessible
- The unused nets, which are functional pins, will have accessibility, except for the Jtag nets (short-circuit detection)
- All the connector pins should be accessible

6.2.1.3 Access choice

Following priority order is to be maintained while accessing the nets:

- Test points
- Connection points: through-hole component, connectors
- Vias

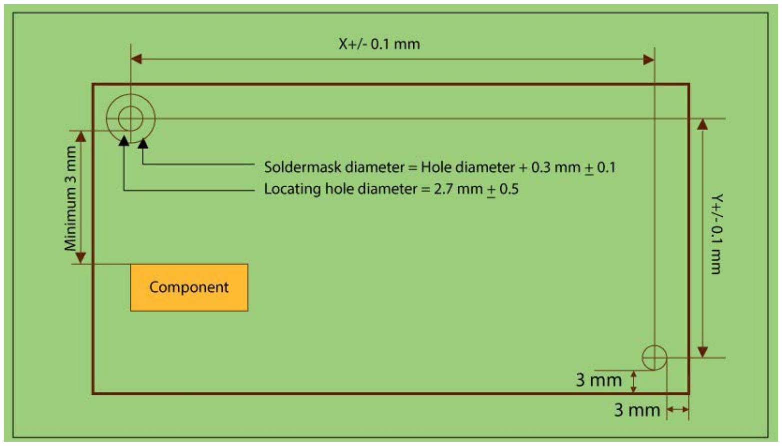
6.2.1.4 Board clamping or locating points

Depending on the board size, two or three locating points are required.

- At least two in the bigger diagonal
- Three, if the board side is greater than 400 mm
- More than 3 for the larger board,
- Include two locating points for each rigid part of the flex-rigid boards

6.2.1.5 Definition of holes

- Preferrable hole diameter is 2.7 mm with ±0.05 tolerance
- Hole locating tolerance is ±0.1 mm
- The solder mask blank is 0.3 mm around the locating holes
- Minimum clearance area:
  - 3 mm between the locating holes and the board edge
  - 3 mm between the edge of the locating holes and each physical element of the board, including the external copper nets
- The locating holes will not be metalized and shall never be used for other purposes



Specifications required to define holes

6.2.1.6 Test point minimum dimensions

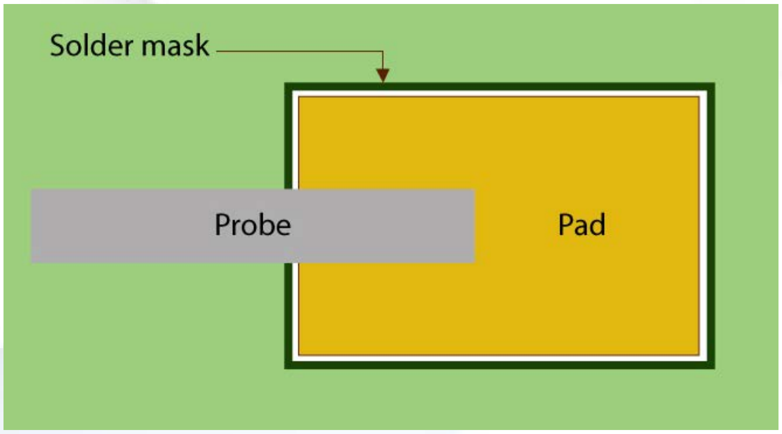
	Test point diameter	Reliability
	1 mm	Highest
Minimum value recommended for TOP side (component side)	0.9 mm	↓
Minimum value recommended for BOTTOM side	0.7 mm	↓
By default for BOTTOM side (Prohibited on the TOP side)	0.6 mm	Lowest

6.2.1.7 Minimum test point size according to board dimensions

Board length and width in mm	<75	75-150	150-300	>300
Recommended test pad diameter	0.7 mm	0.7 mm	0.7 mm	0.9 mm
Minimum accepted test pad diameter	0.6 mm	0.6 mm	0.6 mm	0.7 mm

6.2.1.8 Solder mask and test point

There must not be a solder mask on the test point. The solder mask can not be placed tangent to the test point.



Solder mask should not be on test point

6.2.1.9 Distance between the test points

The distance between the interaxis test points decides the kind of nail used to manufacture the test fixture. There are three types of nails, with different kinds of heads according to the contacts required: point, tulip, and diamond.

6.2.1.10 Nail choice

The choice of the nails determines the cost and reliability. Nails of 1.27mm diameter are used nails when all other possibilities can not be executed.

Nail	Cost factor	Reliability
2.54 mm (100 mils)	1	Highest
1.905 mm (75 mils)	2	↓
1.27 mm (50 mils)	4	Lowest

6.2.1.11 Minimum interaxis

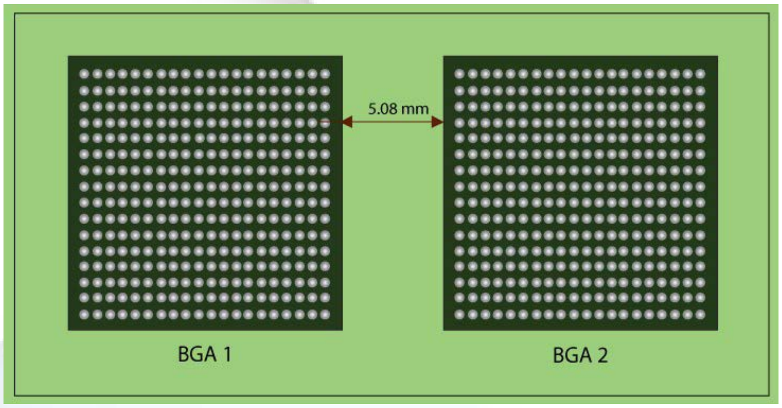
	Minimum interaxis	Type of nail
Minimum recommended value	2 .1 mm	2.54 mm
Minimum recommended value	1.7 mm	1.905 mm
TOP side prohibited by default	1.25 mm	1.27mm

6.2.1.12 Distance between the test point and the component edge

Component type	Distance
Height < 2 mm	0.7 mm
Height > 4 mm	1.91 mm
Board edge	0.9 mm
Hole edge	0.9 mm
Locating edge	3 mm

6.2.1.13 Test point under a BGA component

With a significant amount of pressure, exerted by the nails under BGA components, there is a high risk of breaking the solder joints. Hence, it is crucial to avoid placing a test point under a BGA component. At least, leave a 5.08 mm (200 mils) gap on the four sides of the BGA to allow the placement of the upper contrast with a 3 mm diameter. This clearance is common if two BGA components are placed side by side.



Separating distance between two BGAs

6.2.1.14 Test point density

Distribute the test points on the entire surface to mitigate distortions and mechanic warpage. The test point density should be lower on the board edge with respect to the average board density.

6.2.1.15 Precautions

Check and ensure that the access to the test points is not covered by a connector or by mechanical parts mounted after manufacturing and before the in-circuit test.

6.2.1.16 Accuracy measurement

Incorporate a doubled test point on either side of the component or 4-wire measurement for the components such as resistors below 100 Ohms, and capacitors with an accuracy of 0.1% or with lower functional value.

6.2.1.17 Access to the board power supply

For each power supply, four test points are scheduled to supply a current lower than 1 A. An additional test point can be added to give a rise of 0.5A in the current. It is feasible to schedule as many test points as possible according to the powers on the connector of the board.

6.2.1.18 Wire rework

It is significant to keep in mind that while executing wire rework test points must be excluded from any kind of wiring connection.

6.2.2 ICT benefits

- Used for large production volume
- Provides coverage up to 90%
- Accurate; free from human error

6.2.3 ICT drawbacks

- Not suitable for small volume production
- Voids or inadequate solder masks can not be detected
- Expensive; technologies like test jigs add to more costs

6.2.4 What ICT checks and verifies

- Incorrectly inserted component
- Wrong component
- Improper orientation of diodes, transistors or ICs
- Short circuits and open circuits

7. Other testing methods

7.1 Functional testing

FCT is implemented for quality control and ensures the intended operation of a device. The test parameters are provided by the designers depending on the design. This technique often incorporates simple switch-on/switch-off tests, and sometimes it requires complex software and precise protocols. Functional testing directly checks the board's function in real environmental conditions.

7.1.1 FCT benefits

- Low cost
- Versatile and can be customized as per the design
- Doesn't impact the lifespan of the board, unlike other tests that exert excessive stress on it

7.1.2 FCT drawbacks

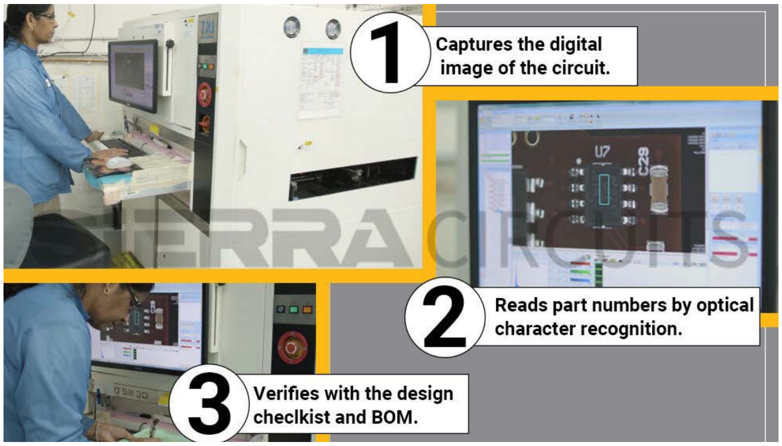
- Requires experienced technicians

7.1.3 What FCT checks and verifies

- Checks the entire assembly
- Verifies boards' functions in different working conditions

7.2 Automated optical inspection

AOI incorporates 2D or 3D cameras that click high-resolution images and verify the schematics. Here, various surface feature defects such as stains, scratches, and open and short circuits are checked. It also scans for missing, incorrect, and wrongly placed components. Automated optical inspection is used with another type of testing method to obtain the correct results. For instance, AOI with the flying probe, and AOI with the in-circuit test. It can be included directly on the production line to prevent any premature board failure.



Automated optical inspection of PCB

7.2.1 How AOI works

1.Capture images: Automated optical inspection incorporates an advanced, expensive camera. These cameras range from extended graphics array (XGA) to multi-megapixels high-resolution sensors. After component placement and soldering, the board is passed through the AOI device. The camera captures multiple images. Successful assembly creates a golden standard. In the subsequent production runs, AOI scans and compares new images with the existing golden standard.

Machine vision camera used in AOI

a. Optical system resolution or imaging resolution is critical. It determines the extent of the details visible to the AOI. The resolution is influenced by inspection speed, accuracy, and size of the smallest component used on the circuit board. The camera's sensor, lens, and the distance from the board are also key factors in determining imaging resolution.
b. Field of view (FOV) is defined by the resolution of the camera sensor. It is the area of the board that can be covered by a single image.

2. Proper illumination for visual inspection: It is paramount to scan and inspect various parts. Different kinds of lights such as incandescent, fluorescent, infrared (IR), and ultraviolet (UV) are involved in AOI. New technologies incorporate LEDs as the light source to procure an even and consistent form of lighting. The lighting system consists of red, white, green, and blue LEDs in a configurable lighting module. These LEDs provide an even and consistent form of lighting.

7.2.2 AOI benefits

- Fatal defects can be detected with accuracy
- Consistent approach

7.2.3 AOI drawbacks

- Only surface defects can be detected
- Time-consuming process
- Setups are subject to change based on design
- Detection based on the database is not always 100% accurate

7.2.4 What AOI checks and verifies

- Lifted and misplaced component
- Open circuits
- Solder shorts
- Insufficient or excess solder

7.3 Burn-in testing

The burn-in testing is an early check-up of the board to prevent dangerous failures after executing fabrication. The method involves exceeding the specified operating limits to trigger the failures. This is an efficient way to detect the maximum operational rating of the board.
Here, the various operating conditions refer to voltage, current, temperature, operating frequency, power, and other factors pertinent to the design.

7.3.1 Burn-in test benefits

- Increases product reliability
- Verifies the functionality of the board concerning the ambient conditions

7.3.2 Burn-in test drawbacks

- The exerted stress beyond the rating can reduce the board's lifespan.
- The process involves more time as well as more effort.

7.3.3 What burn-in test checks and verifies

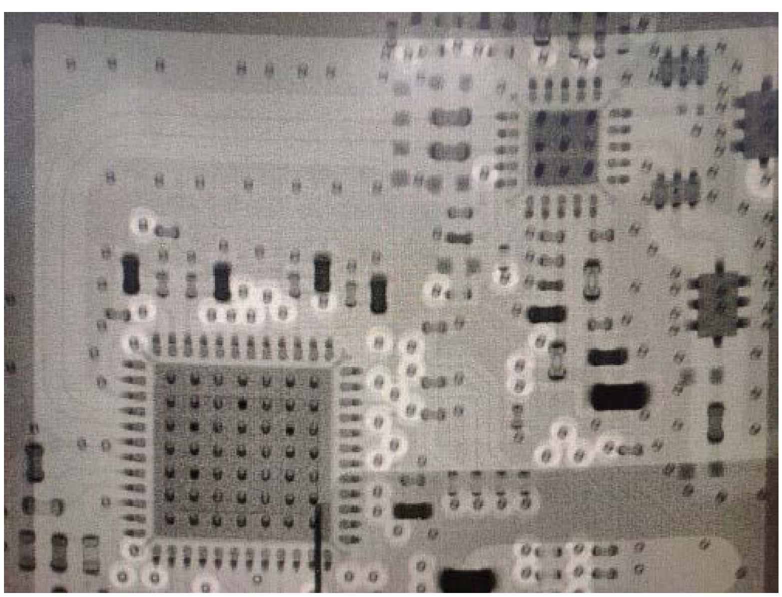
- Examines the reliability of the specified parameters of the components

7.4 Automated X-ray inspection

Automated X-ray inspection (AXI) detects errors in hidden components, solder connections, BGA packages, internal traces, and barrels with the help of X-ray imaging. The machine's software creates real-time 2D images of components and scans the defects in a non-destructive way.

7.4.1 AXI process flow

1. The PCB, under test, is exposed to the X-ray radiation generated by a tube.
2. A detector can capture X-rays passing through the sample board and transform them into a visual representation. Generally, heavy materials absorb more radiation than lighter ones. On X-ray scanned images, heavy materials appear darker whereas lighter materials are more translucent to the radiation. In the case of PCBA, solder joints are made of heavy materials, and other parts such as packages, silicon ICs, and component leads are made of light materials. Hence, high-quality solder joints look darker on images than other parts.



X-ray scanned image of PCB

3. X-ray inspection systems should have enough magnification to ensure the correct information during defect analysis.

X-ray inspection machine at Sierra Circuits

7.4.2 AXI benefits

- An X-ray machine can easily check the internal layers from the top of the board.

7.4.3 AXI drawbacks

- This requires experienced and skilled technicians.
- The process demands more labor and cost.

7.4.4 What AXI checks and verifies

- Misalignment of components
- Solder void
- Solder quality
- BGA open and short connections

7.5 Visual inspection

Here a technician inspects the circuit board with bare eyes or by using a microscope. This method can determine deformed surface features, coating quality, the position of vias, the unveiled components' alignment, the absence of components, and other glitches.

Visual inspection of PCB

7.5.1 Visual inspection benefits

- Easy and basic method

7.5.2 Visual inspection drawbacks

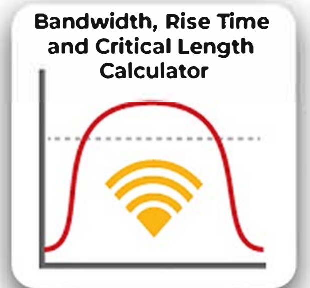
- Subject to human errors
- Minor and invisible defects can not be detected

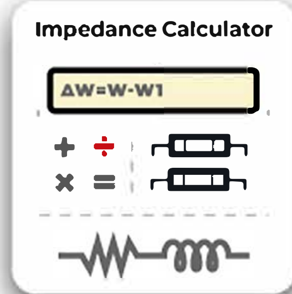
7.5.3 What visual inspection checks and verifies

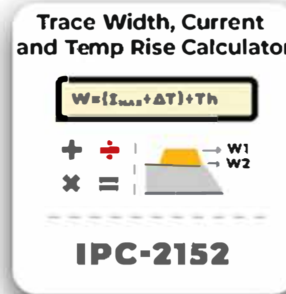
- Missing component
- Contaminated solder paste
- Solder joint reflow
- Warpage issues

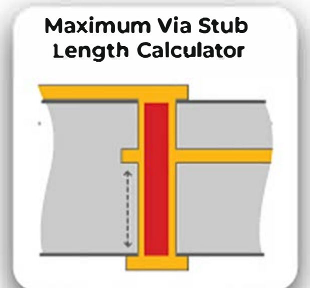
Design for testing is fundamentally a reviewing process to find and fix any manufacturing errors and eliminates excess processes. Without this periodic testing, PCB manufacturers can face serious production errors. DFT ensures fabricating boards with high yields and minimum turnaround times.

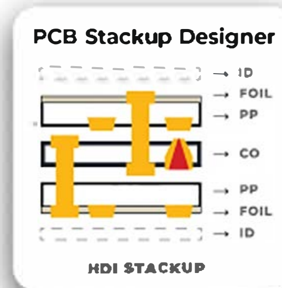
The Designer's Toolkit

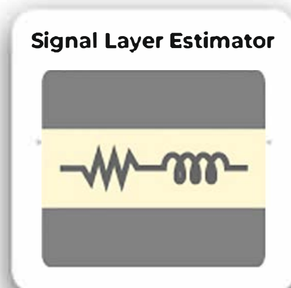


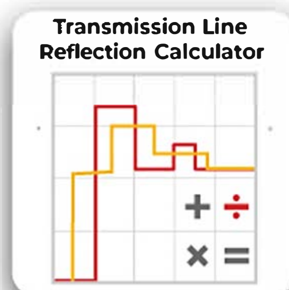


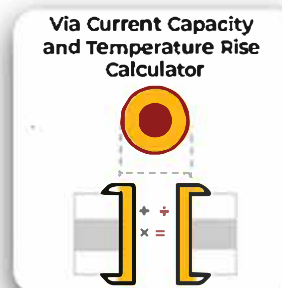


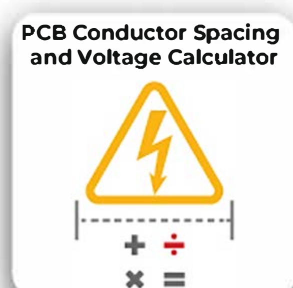


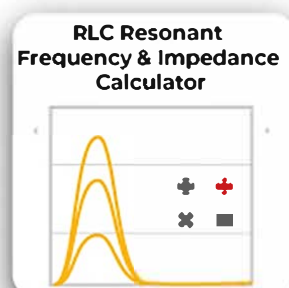


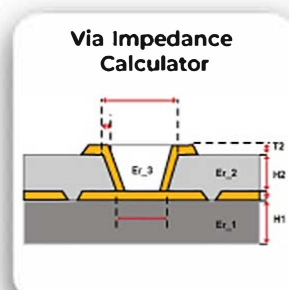


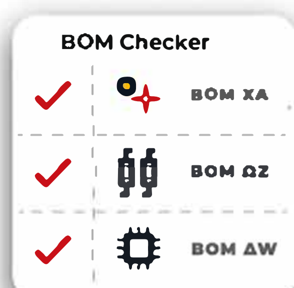












TRY TOOLS

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